

Comparative Analysis of Dimensionality Reduction Techniques for Network Intrusion Detection

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Outline

- 1 Problem Statement
- 2 Objectives
- 3 Experimental Design
- 4 Classifiers Overview
- 5 Dim Reduction Techniques
- 6 Results
- 7 Best Per Classifier
- 8 Conclusion
- 9 Future Work
- 10 References

Problem Statement

The Problem

Network IDS datasets have high dimensionality (118-197 features) causing:

- Slow training time
- Overfitting
- Reduced model accuracy

Research Questions

- 1 Which dim reduction method (PCA / Autoencoder / Mutual Info) works best for each classifier?
- 2 Does the answer change across different datasets?

Our Approach: 5 classifiers $\times$ 4 reduction methods $\times$ 3 datasets = 60 experiments

Objectives

- ❶ Implement **3 dim reduction techniques**: PCA, Autoencoder, Mutual Information
- ❷ Implement **5 classifiers**: LSTM, J48, PART, Random Forest, Naive Bayes
- ❸ Test on **3 datasets**: NSL-KDD, UNSW-NB15, CIC-IDS-2017
- ❹ Compare using **Accuracy, Training Time**
- ❺ Identify best reduction method for each classifier

Datasets	NSL-KDD (118), UNSW-NB15 (197), CIC-IDS (78)
Dim Reduction	None (Raw), PCA, Autoencoder, Mutual Info
Classifiers	LSTM, J48, PART, Random Forest, Naive Bayes
Parameters	Epochs=20, Batch=850, Dropout=0.6, Trees=100

Preprocessing: One-hot encoding + Standard scaling

Five Classifiers Used

Classifier	Type	Key Characteristic
LSTM	Recurrent Neural Network	Designed for sequential data
J48	Decision Tree	C4.5 algorithm, interpretable
PART	Rule-based	Partial decision trees
Random Forest	Ensemble	100 trees, bagging
Naive Bayes	Probabilistic	Assumes feature independence

Three Dimensionality Reduction Techniques

Technique	How it Works
PCA	Finds directions of maximum variance, keeps top components
Autoencoder	Neural network that learns to compress data to 64 features
Mutual Information	Selects features with highest information gain

Parameters: PCA: 30 components | AE: bottleneck=64, 20 epochs | MI: top 30-40 features

Results on NSL-KDD Dataset

Dim Reduction	LSTM	J48	PART	RF	NB
None (Raw)	77.91%	82.95%	79.29%	78.71%	51.36%
PCA	78.02%	81.31%	78.73%	77.96%	79.96%
Autoencoder	77.81%	80.82%	78.21%	77.68%	66.47%
Mutual Info	76.96%	76.79%	80.03%	78.17%	76.33%

Findings: J48 + Raw = best (82.95%). PCA boosted NB by 28.6%.

Results on UNSW-NB15 Dataset

Dim Reduction	LSTM	J48	PART	RF	NB
None (Raw)	85.60%	86.36%	81.68%	87.11%	55.59%
PCA	84.26%	85.39%	80.06%	85.14%	55.94%
Autoencoder	82.57%	85.02%	77.58%	84.92%	49.05%
Mutual Info	85.22%	86.23%	81.19%	87.02%	75.87%

Findings: RF + Raw = best (87.11%). MI boosted NB by 20.3%.

Results on CIC-IDS-2017 Dataset

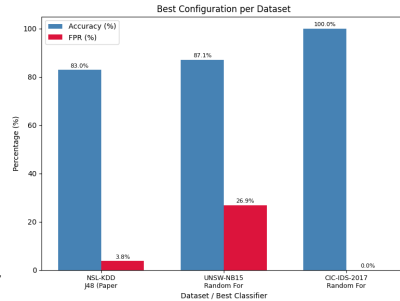
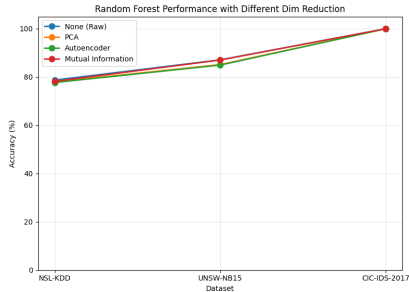
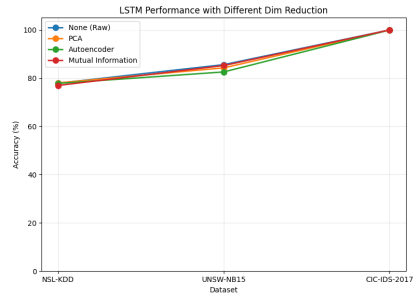
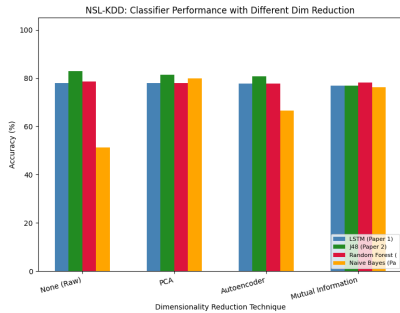
Dim Reduction	LSTM	J48	PART	RF	NB
None (Raw)	99.95%	99.99%	99.93%	100.00%	98.75%
PCA	99.94%	99.96%	99.76%	99.97%	85.28%
Autoencoder	99.90%	99.94%	99.60%	99.98%	94.58%
Mutual Info	99.92%	99.98%	99.93%	99.99%	97.82%

Findings: RF + Raw = 100% accuracy. Dataset is very easy.

Best Dim Reduction for Each Classifier

Classifier	NSL-KDD	UNSW-NB15	CIC-IDS
LSTM	PCA (78.02%)	Raw (85.60%)	Raw (99.95%)
J48	Raw (82.95%)	Raw (86.36%)	Raw (99.99%)
PART	MI (80.03%)	Raw (81.68%)	Raw (99.93%)
Random Forest	Raw (78.71%)	Raw (87.11%)	Raw (100%)
Naive Bayes	PCA (79.96%)	MI (75.87%)	Raw (98.75%)

Key Insight: Raw features work best for 4 out of 5 classifiers. Naive Bayes is the only one that needs reduction.



Key Findings from Comparative Study

Finding 1: Raw Features Work Best for Tree-Based Classifiers

J48 and Random Forest achieved best accuracy with raw features on all three datasets. These classifiers handle high-dimensional data naturally.

Finding 2: Naive Bayes Needs Reduction the Most

PCA improved NB from 51.36% to 79.96% on NSL-KDD (+28.6%). Mutual Info improved NB from 55.59% to 75.87% on UNSW-NB15 (+20.3%).

Finding 3: No Single Reduction Method Works for All Classifiers

LSTM: Best with Raw/PCA | J48/RF: Best with Raw | PART: Best with MI | NB: Best with PCA/MI

Training Time Comparison (UNSW-NB15)

Classifier	Raw	PCA	MI
J48	5.11s	12.29s	212.06s
Random Forest	23.72s	90.82s	237.28s
LSTM	96.04s	91.35s	301.71s

Observation: Mutual Information is very slow on large datasets (197 features, 175k samples).

Conclusion

- ❶ Best reduction method **depends on the classifier**
- ❷ Raw features work best for tree-based classifiers (J48, RF)
- ❸ Naive Bayes needs reduction most (improves by 20-28%)
- ❹ Random Forest achieved **100% accuracy** on CIC-IDS-2017

Future Work

- Test more reduction methods (LDA, t-SNE, Random Projection)
- Test more classifiers (XGBoost, SVM, KNN)
- Parameter tuning for each method
- Test on newer datasets (CIC-IDS-2018, TON-IoT)

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Thank You